CHAPTER NINE

EXERCISE 1

(a) $40° + 80° + a = 180°$ (∠s on a str ln)
$\therefore a + 120° = 180°$
$\therefore a = 60°$

$b = 70°$ (vert opp ∠s)

$c + 120° + 90° + 100° = 360°$ (∠s round a pt)
$\therefore c + 310° = 360°$
$\therefore c = 50°$

(b) $a = 62°$ (corresp ∠s ; AB || CD)
$b = 130°$ (alt ∠s ; RS || PQ)
$c + 45° = 180°$ (alt ∠s ; MN || KL)
$\therefore c = 135°$

$d = 140°$ (corresp ∠s ; AB || CD)
$e = 140°$ (vert opp ∠s)

$f = 42°$ (alt ∠s ; AB || CD)
$g = 42°$ (corresp ∠s ; EF || GH)
$h + 42° = 180°$ (co-int ∠s ; AB || CD)
$\therefore h = 138°$

$i + 160° = 180°$ (∠s on a str ln)
$\therefore i = 20°$

$j + 20° = 180°$ (co-int ∠s ; OH || AG)
$\therefore j = 160°$

$k + 30° = 180°$ (co-int ∠s ; EF || OH)
$\therefore k = 150°$

$l + 160° + 150° = 360°$ (∠s round a pt)
$\therefore l = 50°$

$m + 30° = 50°$ (alt ∠s ; EB || AO)
$\therefore m = 20°$

(c) $a + 60° + 70° = 180°$ ($\angle$ s of Δ)
$\therefore a = 50°$

$b = 52° + 41°$ (ext $\angle$ of Δ)
$\therefore b = 93°$

$c + 65° = 140°$ (ext $\angle$ of Δ)
$\therefore c = 75°$
$d = 68°$ ($\angle$s opp = sides)
$e = 44°$ ($\angle$ s of Δ)

$f + g + 100° = 180°$ ($\angle$ s of Δ)
but $f = g$ ($\angle$s opp = sides)
$\therefore f = g = 40°$

$h = 55°$ ($\angle$s opp = sides)
$i = 110°$ (ext $\angle$ of Δ)

(d) (1) $\hat{F}_1 = 50°$ (alt $\angle$s ; AB $\parallel$ CD)
$\hat{E}_2 = 50°$ ($\angle$s opp = sides)

(2) $\hat{P}_2 = 80°$ ($\angle$ s of Δ)
and $\hat{F}_2 = 80°$ (vert opp $\angle$s)
$\therefore \hat{P}_2 = \hat{F}_2$
$\therefore$ EP $\parallel$ RS (alt $\angle$s =)

(3) $\hat{S}_1 = 50°$ ($\angle$ s of Δ)
but $\hat{P}_3 = 50°$ (given)
$\therefore$ PF = SF (sides opp = $\angle$ s)

(e) (1) $6x + 50° + x + 10° = 180°$ (co-int $\angle$s ; RU $\parallel$ SQ)
$\therefore 7x + 40° = 180°$
$\therefore 7x = 140°$
$\therefore x = 20°$

$y = x + 10°$ (corresp $\angle$s ; RU $\parallel$ SQ)
$\therefore y = 30°$

$\hat{R}_1 = 30°$ ($\angle$s opp = sides)
$\therefore z = 180° - (30° + 30°)$ ($\angle$ s of Δ)
$\therefore z = 120°$

(2) $\hat{R}_3 = 2x - 10°$ (given)
$\therefore \hat{R}_3 = 30°$
$\therefore \hat{R}_2 = 30°$ (vert opp $\angle$s)
but $\hat{U}_1 = y = 30°$ (proven)
$\therefore \hat{R}_2 = \hat{U}_1$
$\therefore$ PQ $\parallel$ RT (alt $\angle$s =)

(f) (1) In ΔWXY and ΔWZY:

$\hat{W}_1 = \hat{W}_2$	(given)
$WY = WY$	(common)
$WX = WZ$	(given)
$\therefore \Delta WXY \equiv \Delta WZY$	(S∠S)

(2) In ΔABC and ΔAED:

$AC = AD$	(sides opp = ∠ s)
$AB = AE$	(given)
$BC = DE$	(given)
$\therefore \Delta ABC \equiv \Delta AED$	(SSS)

(3) In ΔPQR and ΔSTR:

$PQ = ST$	(given)
$\hat{R}_1 = \hat{R}_2$	(vert opp ∠s)
$\hat{P} = \hat{T}$	(alt ∠s ; QP ∥ ST)
$\therefore \Delta PQR \equiv \Delta STR$	(∠∠S)

(4) $\hat{L}_1 = 180° - (37° + 106°)$ (∠ s of Δ)

$\therefore \hat{L}_1 = 37°$

but $\hat{K}_1 = 37°$ (given)

$\therefore KN = LN$ (sides opp = ∠ s)

In ΔKPN and ΔLMN:

$KN = LN$	(proven)
$KP = LM$	(given)
$\hat{P} = \hat{M} = 90°$	(given)
$\therefore \Delta KPN \equiv \Delta LMN$	(RHS)

(g) (1) In ΔABE and ΔACD:

$\hat{A} = \hat{A}$	(common)
$AE = AD$	(given)
$\hat{B}_1 = \hat{C}_1$	(given)
$\therefore \Delta ABE \equiv \Delta ACD$	(∠∠S)
$\therefore AB = AC$	(≡ Δs)

(2) $\hat{B}_1 + \hat{B}_2 = \hat{C}_1 + \hat{C}_2$ (∠s opp = sides)

but $\hat{B}_1 = \hat{C}_1$ (given)

$\therefore \hat{B}_2 = \hat{C}_2$

$\therefore BO = OC$ (sides opp = ∠ s)

(3) In ΔDOB and ΔEOC:

$\hat{B}_1 = \hat{C}_1$	(given)
$\hat{O}_1 = \hat{O}_2$	(vert opp ∠s)
$BO = OC$	(proven)
$\therefore \Delta DOB \equiv \Delta EOC$	(∠∠S)

(h) (1) In ΔABC and ΔAED:

$\hat{A} = \hat{A}$ (common)

$\hat{B} = \hat{E}_1$ (given)

$\hat{C} = \hat{D}_1$ (3rd $\angle$ of Δ)

$\therefore \Delta ABC \,|||\, \Delta AED$ ($\angle\angle\angle$)

(2) In ΔPQR and ΔTSR:

$\hat{R}_1 = \hat{R}_2$ (vert opp $\angle$s)

$\hat{P} = \hat{T}$ (alt $\angle$s ; PQ $\parallel$ ST)

$\hat{Q} = \hat{S}$ (alt $\angle$s ; PQ $\parallel$ ST)

$\therefore \Delta PQR \,|||\, \Delta TSR$ ($\angle\angle\angle$)

(3) In ΔABC and ΔDEF:

$$\frac{AB}{DE} = \frac{5}{3}$$

$$\frac{AC}{DF} = \frac{10}{6} = \frac{5}{3}$$

$$\frac{BC}{EF} = \frac{7{,}5}{4{,}5} = \frac{5}{3}$$

$$\therefore \frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$$

$\therefore \Delta ABC \,|||\, \Delta DEF$ (sides of Δs in prop)

(4) In ΔMRP and ΔQNP:

$$\frac{MR}{QN} = \frac{12}{9} = \frac{4}{3}$$

$$\frac{RP}{NP} = \frac{20}{15} = \frac{4}{3}$$

$$\frac{MP}{QP} = \frac{16}{12} = \frac{4}{3}$$

$$\therefore \frac{MR}{QN} = \frac{RP}{NP} = \frac{MP}{QP}$$

$\therefore \Delta MRP \,|||\, \Delta QNP$ (sides of Δs in prop)

(i) (1) $\dfrac{BC}{ED} = \dfrac{AC}{FD}$ ($|||$ Δs)

$$\therefore \frac{BC}{16} = \frac{27}{12}$$

$\therefore BC = 36$ units

(2) $\hat{B} = \hat{E}_2$ ($|||$ Δs)

$\therefore ED \parallel BC$ (alt $\angle$s =)

(j) (1) $AB^2 = AE^2 + BE^2$ (Pythagoras)

$\therefore AB^2 = (12)^2 + (5)^2$

$\therefore AB = 13$ cm

(2) In ΔABE and ΔADC:

$\hat{A} = \hat{A}$ (common)

$\hat{E} = \hat{C} = 90°$ (given)

$\hat{B}_1 = \hat{D}$ (3rd $\angle$ of Δ)

$\therefore \Delta ABE \,|||\, \Delta ADC$ ($\angle\angle\angle$)

(3)(i) $\dfrac{CD}{BE} = \dfrac{AD}{AB}$ ($|||\ \Delta s$)

$\therefore \dfrac{CD}{5} = \dfrac{26}{13}$

$\therefore CD = 10$ cm

(ii) $\dfrac{AC}{AE} = \dfrac{AD}{AB}$ ($|||\ \Delta s$)

$\therefore \dfrac{AC}{12} = \dfrac{26}{13}$

$\therefore AC = 24$ cm

but $AB = 13$ cm

$\therefore BC = 11$ cm (proven)

(k) (1)(i) $QR^2 = QU^2 + UR^2$ (Pythagoras)

$\therefore QR^2 = (30)^2 + (16)^2$

$\therefore QR = 34$ m

(ii) $PT^2 = PS^2 - TS^2$ (Pythagoras)

$\therefore PT^2 = (51)^2 - (24)^2$

$\therefore PT = 45$ m

(2) In ΔPTS and ΔQUR:

$\dfrac{PT}{QU} = \dfrac{45}{30} = \dfrac{3}{2}$

$\dfrac{TS}{US} = \dfrac{24}{16} = \dfrac{3}{2}$

$\dfrac{PS}{QR} = \dfrac{51}{34} = \dfrac{3}{2}$

$\therefore \Delta PTS \,|||\, \Delta QUR$ (sides of Δs in prop)

$\therefore \hat{P} = \hat{Q}_1$ ($|||\ \Delta s$)

$\therefore PT \parallel QU$ (corresp $\angle$s =)

EXERCISE 2

(a) (1) $AP = PB$ and $AQ = QC$ (given)

$\therefore PQ \parallel BC$ (midpt thm)

$\therefore \hat{P}_1 = 69°$ (corresp $\angle$s ; $PQ \parallel BC$)

(2) $PQ = \dfrac{1}{2}BC$ (midpt thm)

$\therefore PQ = 4$ cm

(b) (1) $AR = AP$ (given)
$PQ \parallel AB$ (given)
$\therefore BR = BQ$ (line through midpt $\parallel$ 2nd side)
$\therefore BR = 12$ m

(2) $AB = \frac{1}{2}PQ$ (midpt thm)
$\therefore PQ = 2AB$
$\therefore PQ = 20$ m

(c) (1) $DP = PF$ and $EQ = QF$ (given)
$\therefore DE \parallel PQ$ and $PQ = \frac{1}{2}DE$ (midpt thm)
$\therefore PQ = 1{,}5$ m

(2) $DE \parallel PQ$ (midpt thm)
$\therefore \hat{P}_1 = 75°$ (alt $\angle$s ; $DE \parallel PQ$)

(3)(i) $AE = EC$ (given)
$DE \parallel BC$ (proven)
$\therefore AD = DB$ (line through midpt $\parallel$ 2nd side)
$\therefore AD = 3$ cm

(ii) $DE = \frac{1}{2}BC$ (midpt thm)
$\therefore BC = 2DE$
$\therefore BC = 6$ cm

(iii) $AB = 6$ cm (given)
$BC = 6$ cm (proven)
$\therefore \hat{A} = \hat{C} = 55°$ ($\angle$s opp = sides)
$\therefore \hat{B} = 180° - (55° + 55°)$ ($\angle$s of Δ)
$\therefore \hat{B} = 70°$

(d) (1)(i) $TU = UR$ and $PQ \parallel SV$ (given)
$\therefore QV = VR$ (line through midpt $\parallel$ 2nd side)
but $QV = 6$ m (given)
$\therefore QR = 12$ m

(ii) $UV = \frac{1}{2}TQ$ (midpt thm)
$\therefore UV = 2$ m

(2) $TU = UR$ and $PQ \parallel SV$ (given)
$\therefore PS = SR$ (line through midpt $\parallel$ 2nd side)
and $SU = \frac{1}{2}PT$ (midpt thm)
$\therefore PT = 8$ m
$PQ = PT + TQ = 8 + 4 = 12$ m
and $QR = QV + VR = 6 + 6 = 12$ m
$\therefore PQ = QR$
$\therefore \Delta PQR$ is isosceles

(3) $\hat{P} = Q\hat{R}P$ (∠s opp = sides)

$\therefore \hat{P} = \frac{180° - 110°}{2}$

$\therefore \hat{P} = 35°$

$\hat{S}_2 = 35°$ (corresp ∠s ; PQ || SV)

(e) (1) AF = FE and BG = GE (given)

$\therefore FG = \frac{1}{2}AB$ (midpt thm)

$\therefore AB = 2FG$

$\therefore AB = 14$ units

(2) $EC^2 = AE^2 - AC^2$ (Pythagoras)

$\therefore EC^2 = (26)^2 - (24)^2$

$\therefore EC^2 = 100$

$\therefore EC = 10$ units

(3) FG || AB (midpt thm)

EG = BG (given)

$\therefore ED = DC$ (line through midpt || 2nd side)

$\therefore ED = 5$ units

(f) AE = ED and AB || EF (given)

$\therefore BF = FD$ (line through midpt || 2nd side)

but DG = GC (given)

$\therefore BC \parallel FG$ (midpt thm)

(g) AP = PO and AB || PQ (given)

$\therefore BQ = QO$ (line through midpt || 2nd side)

and BC || QR (given)

$\therefore OR = RC$ (line through midpt || 2nd side)

$\therefore PR = \frac{1}{2}AC$ (midpt thm)

(h) (1)(i) $\hat{Y}_1 = \hat{M}$ (given)

$\therefore XY \parallel KM$ (corresp ∠s =)

and LY = YM (given)

$\therefore KX = XL$ (line through midpt || 2nd side)

(ii) KX = XL (proven)

KW = WM (given)

$\therefore XW = \frac{1}{2}LM$ (midpt thm)

but LY = YM (given)

$\therefore XW = YM$

(2) $\hat{Y}_1 = 40°$ (given)

$\therefore \hat{M} = 40°$ (given)

but XW || LM (midpt thm)

$\therefore \hat{W}_2 = 180° - 40°$ (co-int ∠s ; XW || LM)

$\therefore \hat{W}_2 = 140°$

(i) (1) $\hat{A}_1 = \hat{C}_2$ (given)

$\therefore$ AB $\parallel$ EC (alt $\angle$s =)

and BC = CD (given)

$\therefore$ AE = ED (line through midpt $\parallel$ 2nd side)

(2) AF = FC and EG = GC (given)

$\therefore FG = \frac{1}{2}AE$ (midpt thm)

but AE = ED (proven)

$\therefore FG = \frac{1}{4}AD$

(j) (1) PA = AQ and PB = BR (given)

and AB $\parallel$ QR (midpt thm)

but DE $\parallel$ QR (given)

$\therefore$ AB $\parallel$ DE

and AD = DC (given)

$\therefore$ BE = EC (line through midpt $\parallel$ 2nd side)

(2) $DE = \frac{1}{2}AB$ (midpt thm)

and QR = 2AB (midpt thm)

but AB = 2DE

$\therefore$ QR = 2(2DE)

$\therefore$ QR = 4DE

(k) (1) AF = FD and AE = EB (given)

$\therefore$ FE $\parallel$ BD (midpt thm)

but FE $\parallel$ HI (given)

$\therefore$ BD $\parallel$ HI

and BH = HC (given)

$\therefore$ DI = IC (line through midpt $\parallel$ 2nd side)

(2) $EF = \frac{1}{2}DB$ (midpt thm)

and $HI = \frac{1}{2}BD$ (midpt thm)

$\therefore$ EF = HI

In ΔEFG and ΔIHG:

$\hat{G}_1 = \hat{G}_2$ (vert opp $\angle$s)

$E\hat{F}G = G\hat{H}I$ (alt $\angle$s ; AB $\parallel$ CD)

EF = HI (proven)

$\therefore \Delta EFG \equiv \Delta IHG$ ($\angle\angle$S)

$\therefore$ FG = GH

EXERCISE 3

(a) $\hat{U} + \hat{A} + \hat{D} + \hat{Q} = 360°$ (∠s of quad)
$\therefore \hat{U} + 90° + 70° + 60° = 360°$
$\therefore \hat{U} + 220° = 360°$
$\therefore \hat{U} = 140°$

(b) (1) $\hat{P} + \hat{Q} + \hat{R} + \hat{S} = 360°$ (∠s of quad)
$\therefore a - 10° + 3a - 50° + 2a - 20° + a + 20° = 360°$
$\therefore 7a - 60° = 360°$
$\therefore 7a = 420°$
$\therefore a = 60°$

(2) $\hat{P} = a - 10° = 60° - 10° = 50°$
$\hat{Q} = 3a - 50° = 3(60°) - 50° = 130°$
$\therefore \hat{P} + \hat{Q} = 180°$
$\therefore$ PS ∥ QR (co-int ∠s suppl)
$\therefore$ PQRS is a trapezium (one pair opp sides ∥)

(c) (1) AD = BC = 3 m (opp sides of parm)
but BC = PC (given)
$\therefore$ PC = 3 m

(2)(i) $\hat{D} = \hat{B} = 65°$ (opp ∠s of parm)
$\therefore \hat{P}_2 = 65°$ (∠s opp = sides)
$\therefore \hat{C}_1 = 180° - (65° + 65°)$ (∠s of Δ)
$\therefore \hat{C}_1 = 50°$

(ii) AB ∥ DC (opp sides of parm)
$\therefore \hat{P}_2 = 65°$ (alt ∠s ; AB ∥ DC)

(iii) $\hat{A} + 65° = 180°$ (co-int ∠s ; AB ∥ DC)
$\therefore \hat{A} = 115°$

(d) (1) SR = PQ = 4 cm (opp sides of parm)
$\therefore$ TR = 4 cm (SR = TR given)
PS = QT + TR (opp sides of parm)
$\therefore$ PS = 2 + 4
$\therefore$ PS = 6 cm

(2)(i) $\hat{T}_2 = 60°$ (equilateral Δ)
$\therefore \hat{T}_1 = 180° - 60°$ (∠s on a str ln)
$\therefore \hat{T}_1 = 120°$

(ii) $\hat{P} = \hat{R} = 60°$ (opp ∠s of parm)

(iii) QR ∥ PS (opp sides of parm)
$\hat{S}_2 = 60°$ (alt ∠s ; QR ∥ PS)

(e) (1) OP = OR = 25 mm (diags of parm)

(2) PM = 25 mm (PO = PM given)
$\therefore$ AR = 25 mm (opp sides of parm)

(3) $MA = MO + OA$
$\therefore MA = 30 + 30$ (diags of parm)
$\therefore MA = 60$ mm

(f) (1) $AO = 3$ m (diags of parm)
but $AO = OP$ (given)
$\therefore OP = 3$ m

(2)(i) $AO = OC$ (diags of parm)
but $AO = OP$ (given)
$OP = OC$
$\hat{C}_1 = 40°$ (alt ∠s ; AB || DC)
$\therefore \hat{P}_1 = 40°$ (∠s opp = sides)
(ii) $\hat{D}_1 + \hat{O}_1 = 40°$ (ext ∠ of Δ)
$\therefore \hat{D}_1 = 20°$ (∠s opp = sides)
$\therefore \hat{B}_1 = 20°$ (alt ∠s ; AB || DC)

(g) (1) AB || DC (opp sides of parm)
$\hat{C}_1 + \hat{C}_2 + \hat{B} = 180°$ (co-int ∠s ; AB || DC)
$\therefore x + x + 10° + 3x + 20° = 180°$
$\therefore 5x + 30° = 180°$
$\therefore x = 30°$
$\hat{C}_2 = x + 10° = 30° + 10° = 40°$
AD || BC (opp sides of parm)
$\therefore y = 140°$ (co-int ∠s ; AD || BC)

(2) $\hat{C}_1 = \hat{E}_1 = x$ (∠s opp = sides)
but AB || DC (opp sides of parm)
$\therefore \hat{C}_2 = x$ (alt ∠s ; AB || DC)
but $\hat{C}_1 + \hat{C}_2 + \hat{D} = 180°$ (co-int ∠s ; AD || BC)
$\therefore x + x + x + 30° = 180°$
$\therefore 3x + 30° = 180°$
$\therefore x = 50°$
$\hat{D} = x + 30° = 50° + 30° = 80°$
$y + 80° = 180°$ (co-int ∠s ; AB || DC)
$\therefore y = 100°$

(3) $\hat{A} = \hat{C}_1$ (opp ∠s of parm)
$\therefore 2x + 80° = 5x + 50°$
$\therefore 3x = 30°$
$\therefore x = 10°$
$\hat{C}_1 = 5x + 50° = 5(10°) + 50° = 100°$
$\therefore \hat{C}_2 = 80°$ (∠s on a str ln)
DC = AB (opp sides of parm)
$\therefore$ DC = DE
$\therefore y = 80°$ (∠s opp = sides)

(4)

$AB \parallel DC$ (opp sides of parm)
$\hat{A}_1 = 2x$ (alt $\angle$s ; $AB \parallel DC$)
but $AO = OC$ (diags of parm)
and $OC = OE$ (given)
$\therefore AO = OE$
$\therefore \hat{E}_1 = 2x$ ($\angle$s opp = sides)
but $\hat{A}_1 + \hat{O}_1 + \hat{E}_1 = 180°$ ($\angle$s of Δ)
$\therefore 2x + 2x + 5x = 180°$
$\therefore 9x = 180°$
$\therefore x = 20°$
$\hat{E}_1 = 2x = 2(20°) = 40°$
$\therefore y + 25° = 40°$ (ext $\angle$ of Δ)
$\therefore y = 15°$

(h) (1) $OL = 8$ cm (diags of parm)

(2) $KM = 10$ cm (given)
$\therefore KO = 5$ cm (diags of parm)

(3) $KO = OM$ (diags of parm)
$KP \parallel OS$ (given)
$\therefore PS = SM$ (line through midpt $\parallel$ 2nd side)
$\therefore PM = 14$ cm

(i) (1) $AO = OC$ (diags of parm)
$AP = PD$ (diags of parm)
$\therefore OP \parallel CD$ (midpt thm)

(2)(i) $BO = OP$ and $OP = PE$ (diags of parm)
$\therefore BO = OP = PE$
but $BE = BO + OP + PE$
$\therefore 30P = 9$
$\therefore OP = 3$ units

(ii) $CD = 20P$ (midpt thm)
$\therefore CD = 2(3)$
$\therefore CD = 6$ units

(j) (1) $\text{Area of ABCD} = \frac{1}{2}(AB + DC) \times \perp h$
$= \frac{1}{2}(10 + 15) \times 5$
$= 62{,}5 \text{ cm}^2$

(2) $\text{Area of ABCD} = AB \times AP$
$= 8 \times 4$
$= 32 \text{ m}^2$

(3) $AP^2 = AB^2 - BP^2$ (Pythagoras)
$\therefore AP^2 = (13)^2 - (5)^2$
$\therefore AP^2 = 144$
$\therefore AP = 12$ cm

Area of ABCD $= BC \times AP$
$= 25 \times 12$
$= 300$ cm^2

(4) Construction: Draw line DE $\perp$ AB
EB = DC = 15 cm
$DE^2 = AD^2 - AE^2$ (Pythagoras)
$\therefore DE^2 = (10)^2 - (6)^2$
$\therefore DE^2 = 64$
$\therefore DE = 8$ cm

Area of ABCD $= \frac{1}{2}(AB + DC) \times DE$
$= \frac{1}{2}(21 + 15) \times 8$
$= 144$ cm^2

(5) $AB^2 = BE^2 - AE^2$ (Pythagoras)
$\therefore AB^2 = (25)^2 - (15)^2$
$\therefore AB^2 = 400$
$\therefore AB = 20$ cm

Area of ABCD $= \frac{1}{2}(AB + CD) \times AE$
$= \frac{1}{2}(20 + 32) \times 15$
$= 390$ cm^2

(6) $PC^2 = BC^2 - BP^2$ (Pythagoras)
$\therefore PC^2 = (13)^2 - (12)^2$
$\therefore PC^2 = 25$
$\therefore PC = 5$ m
DP = 20 m
$\therefore$ DC = 15 m
Area of ABCD $= DC \times BP$
$= 15 \times 12$
$= 180$ m^2

(k)* (1)(i) BE = 20 cm (opp sides of parm)
$AB^2 = BE^2 - AE^2$ (Pythagoras)
$\therefore AB^2 = (20)^2 - (16)^2$
$\therefore AB^2 = 144$
$\therefore AB = 12$ cm

(ii) $EC^2 = AE^2 + AC^2$ (Pythagoras)
$\therefore EC^2 = (16)^2 + (30)^2$
$\therefore EC^2 = 1\,156$
$\therefore EC = 34$ cm
$EF = \frac{1}{2}EC$ (diags of parm)
$\therefore EF = \frac{1}{2}(34)$
$\therefore EF = 17$ cm

(2)(i) Area of BCDE $= ED \times AE$
$= 18 \times 16$
$= 288\ cm^2$

(ii) Area of AEDC $= \frac{1}{2}(AC + ED) \times AE$
$= \frac{1}{2}(30 + 18) \times 16$
$= 384\ cm^2$

EXERCISE 4

(a)
$a = 90°$ (diags of rhombus)
$b = 65°$ ($\angle$s of Δ)
$c = 65°$ (diags of rhombus)
AD = DC (sides of rhombus)
$d = 65°$ ($\angle$s opp = sides)
$e = 45°$ (diags of square)
$f = 25°$ (ext $\angle$ of Δ)

PM = SM (diags of rect)
$g + h + 42° = 180°$ ($\angle$s of Δ)
but $g = h$ ($\angle$s opp = sides)
$\therefore g = 69°$
and $h = 69°$

$i = 21°$ ($\angle$ of rect)
PQ || SR (opp sides of rect)
$\therefore j = 21°$ (alt $\angle$s ; PQ || SR)
$N\hat{K}L = 90°$ ($\angle$ of square)
$\therefore k = 45°$ (diags of square)

$\therefore l + m + 45° = 180°$ ($\angle$s of Δ)
but $l = m$ ($\angle$s opp = sides)
$\therefore l = 67{,}5°$
and m = 67,5°
$n = 22{,}5°$ (diags of square)

(b) (1) OA = 5 cm (diags of rect)
$\therefore$ DB = 10 cm (diags of rect)

(2) $\hat{A} = 90°$ (∠ of rect)
$AB^2 = DB^2 - AD^2$ (Pythagoras)
$\therefore AB^2 = (10)^2 - (6)^2 = 64$
$\therefore AB = 8$ cm

(c) $TO = 21$ cm (diags of rhombus)
$OS = 20$ cm (diags of rhombus)
$\hat{O}_2 = 90°$ (diags of rhombus)
$TS^2 = TO^2 + OS^2$ (Pythagoras)
$\therefore TS^2 = (21)^2 + (20)^2 = 841$
$\therefore TS = 29$ cm

(d) (1) $A\hat{D}C = 90°$ (∠ of square)
$\therefore \hat{D}_1 = 45°$ (diag of square)
$\therefore \hat{P}_1 = 95°$ (ext ∠ of Δ)

(2) $\hat{D}_1 + \hat{D}_2 = 50°$ (∠s opp = sides)
$\therefore \hat{D}_2 = 5°$

(e) (1) $\hat{D} = \hat{B}_1$ (opp ∠s of parm)
$\therefore \hat{B}_1 = 36°$
$\therefore \hat{B}_2 = 36°$ (diags of rhombus)

(2) $\hat{O}_1 = 90°$ (diags of rhombus)
$\therefore \hat{A}_1 = 54°$ (∠s of Δ)

(f) (1) $\hat{B}_1 + \hat{B}_2 = 90°$ (∠ of rect)
and $\hat{O}_1 = 90°$ (diags of rhombus)
$\therefore \hat{B}_1 + \hat{B}_2 + \hat{O}_1 = 180°$
$\therefore AB \parallel PQ$ (co-int ∠s suppl)

(2)(i) $\hat{D} = 90°$ (∠ of rect)
$\therefore \hat{D}_1 = 40°$
$\therefore \hat{C}_1 = 30°$ (∠s of Δ)

(ii) $AB \parallel DC$ (opp sides of rect)
but $AB \parallel PQ$ (proven)
$\therefore PQ \parallel DC$
$\therefore \hat{P}_2 = 30°$ (alt ∠s ; $PQ \parallel DC$)
$\therefore \hat{P}_1 = 30°$ (diags of rhombus)

(iii) $PC = CQ$ (sides of rhombus)
$\therefore \hat{Q}_2 = 30°$ (∠s opp = sides)

(g) (1)(i) $BC = BD$ (given)
but $BC = CD$ (sides of rhombus)
$\therefore BC = CD = BC$
$\therefore \hat{B}_3 = 60°$ (equilateral Δ)

(ii) $\hat{B}_2 = 60°$ (diags of rhombus)
but $\hat{B}_1 + \hat{B}_2 = 90°$ ($\angle$ of rect)
$\therefore \hat{B}_1 = 30°$

(2) $OD = 1$ m
$\therefore AD = 2$ cm (sides of rhombus)
$AB^2 = AD^2 - BD^2$ (diags of rect)
$\therefore AB^2 = (2)^2 - (1) = 3$ (Pythagoras)
$\therefore AB = \sqrt{3} = 1{,}73$ m

(h) (1) $\hat{P}_1 = 90°$ (diags of rhombus)
$\therefore \hat{O}_1 = 54°$ ($\angle$s of Δ)

(2) $\hat{O}_2 = 54°$ (diags of rhombus)
$\hat{O}_3 = 90°$ ($\angle$ of square)
$\therefore \hat{O}_4 = 162°$ ($\angle$s round a pt)

(3) $EO = OB$ (sides of rhombus)
and $OB = OD$ (sides of square)
$\therefore EO = OD$
$\therefore \hat{E}_1 = \hat{D}_1$ ($\angle$s opp = sides)
but $\hat{E}_1 + \hat{D}_1 + 162° = 180°$ ($\angle$s of Δ)
$\therefore \hat{E}_1 = 9°$

(i) (1) $BP = PC$ (sides opp = $\angle$s)
but $PC = CD = 37$ m (sides of rhombus)
$\therefore BP = 37$ m

(2) $PE = 37$ m (sides of rhombus)
$\therefore BP = PE$ (both = 37 m)
and $BO = OF$ (diags of rhombus)
$\therefore OP \parallel FE$ (midpt thm)

(3) $AB = 74$ m (sides of rhombus)
and $OB = 70$ m (diags of rhombus)
and $\hat{O}_5 = 90°$ (diags of rhombus)
$AO^2 = AB^2 - OB^2$
$\therefore AO^2 = (74)^2 - (70)^2 = 576$
$\therefore AO = 24$ m
$AO = OE$ (diags of rhombus)
$\therefore AE = 48$ m

(j) (1) $\hat{O}_1 = 90°$ (diags of rhombus)
$\hat{C}_1 = x + 12°$ (diags of rhombus)
$\therefore 2x + 18° + x + 12° + 90° = 180°$ ($\angle$s of Δ)
$\therefore 3x = 60°$
$\therefore x = 20°$
$\therefore B_2 = 2(20°) + 18° = 58°$
$y = 58°$ (diags of rhombus)
AD = DC (sides of rhombus)
$z = 32°$ ($\angle$s opp = sides)

(2) $\hat{A} = x$ ($\angle$s opp = sides)
$\therefore x - 21° + x + x = 180°$ ($\angle$s of Δ)
$\therefore 3x = 201°$
$\therefore x = 67°$
OQ = OR (diags of rect)
$\therefore \hat{Q}_1 + \hat{Q}_2 = 67°$ ($\angle$s opp = sides)
but $Q_1 = 67° - 21° = 46°$
$\therefore Q_2 = y = 21°$
$Q_1 + Q_2 + Q_3 = 90°$ ($\angle$ of rect)
$\therefore Q_3 = z = 23°$

(3) $\hat{M}_3 = 45°$ (diags of square)
$\therefore 4x + 3x - 5° + 45° = 180°$ ($\angle$s on a str ln)
$\therefore 7x = 140°$
$\therefore x = 20°$

$\hat{M}_4 = 3(20°) - 5 = 55°$
and $\hat{M}_2 + \hat{M}_3 = 55°$ (diags of rhombus)
but $\hat{M}_3 = 45°$ (diags of square)
$\therefore \hat{M}_2 = y = 10°$

$\hat{O}_1 = 90°$ (diags of square)
$\therefore z = 90° + 10°$ (ext $\angle$ of Δ)
$\therefore z = 100°$

(k) (1) Area of ABCD $= b \times \perp h$
$= 10 \times 7$
$= 70 \text{ cm}^2$

(2) $\therefore$ Area of ABCD $= b \times \perp h$
$= 4 \times 3$
$= 12 \text{ cm}^2$

(3) AD = DC = 12 m (sides of rhombus)
$\therefore$ Area of ABCD $= b \times \perp h$
$= 12 \times 10$
$= 120 \text{ m}^2$

(4) $AD^2 = AP^2 + DP^2$ (Pythagoras)
$\therefore AD^2 = (77)^2 + (36)^2 = 7225$
$\therefore AD = 85$ cm
$\therefore DC = 85$ cm (sides of rhombus)

$\therefore$ Area of ABCD $= b \times \perp h$
$= 85 \times 77$
$= 6\,545$ cm^2

(5) $DC^2 = CP^2 + PD^2$ (Pythagoras)
$\therefore DC^2 = (24)^2 + (7)^2 = 625$
$\therefore DC = 25$ m
but $DC = AD$ (sides of rhombus)
$\therefore AD = 25$ m
$\therefore$ Area of ABCD $= b \times \perp h$
$= 25 \times 24$
$= 600$ m^2

(6) $AD = CD = 17$ cm (sides of rhombus)
$\therefore PD = 8$ cm
$AP^2 = AD^2 - PD^2$ (Pythagoras)
$\therefore AP^2 = (17)^2 - (8)^2 = 225$
$\therefore AP = 15$ cm

Area of ABCD $= b \times \perp h$
$= 17 \times 15$
$= 255$ cm^2

(l)* (1) $AD = DC = 2x + 1$ (sides of rhombus)
$\therefore DP = x + 1$
$AD^2 = AP^2 + DP^2$ (Pythagoras)
$\therefore (2x + 1)^2 = (4)^2 + (x + 1)^2$
$\therefore 4x^2 + 4x + 1 = 16 + x^2 + 2x + 1$
$\therefore 3x^2 + 2x - 16 = 0$
$\therefore (3x + 8)(x - 2) = 0$
$\therefore x = -\frac{8}{3}$ or $x = 2$
N.A.
$\therefore x = 2$ units

(2) $DC = 2(2) + 1 = 5$ units
Area of ABCD $= b \times \perp h$
$= 5 \times 4$
$= 20$ units^2

EXERCISE 5

(a) (1) $\hat{A}_2 = 34°$ (diags of kite)

(2) $\hat{D}_1 = 120°$ (∠s opp main diag of kite)

(3) $\hat{C}_2 = 180° - (120° + 34°)$ (∠s of Δ)
$\therefore \hat{C}_2 = 26°$
$\therefore \hat{D}_2 = 26°$ (alt ∠s ; AC || DF)
$\therefore \hat{D}_3 = 26°$ (diags of kite)

(4) $\hat{O}_2 = 90°$ (diags of kite)
$\hat{E}_1 = 180° - (90° + 26°)$ (∠s of Δ)
$\therefore \hat{E}_1 = 64°$

(5) $\hat{F}_1 + \hat{F}_2 = 360° - 238°$ (∠s round a pt)
$\hat{F}_1 + \hat{F}_2 = 122°$
but $\hat{F}_1 + \hat{F}_2$ (diags of kite)
$\therefore \hat{F}_1 = 61°$

(6) $\hat{O}_4 = 90°$ (diags of kite)
$\therefore \hat{C}_4 = 180° - (90° + 61°)$ (∠s of Δ)
$\therefore \hat{C}_4 = 29°$

(b) (1) $\hat{D}_2 = 45°$ (diags of square)
$\therefore \hat{D}_3 = 45°$ (diags of kite)

(2) $\therefore \hat{O}_1 + \hat{O}_2 = 110°$ (∠s opp main diag of kite)
but $\hat{O}_1 = 90°$ (diags of square)
$\therefore \hat{O}_2 = 20°$

(3) $\hat{T}_2 = 25°$ (∠s of Δ)

(4) $\hat{T}_1 = 25°$ (diags of square)
$\hat{C}_1 + \hat{C}_2 = 90°$ (∠ of square)
$\therefore \hat{C}_3 = 90°$ (∠s on a str ln)
$\therefore \hat{R}_1 = 115°$ (ext ∠ of Δ)

(c) (1) $\hat{F}_2 = 76°$ (∠s opp main diag of kite)

(2) $\hat{B} = 76°$ (opp ∠s of rhombus)
BC = CD (sides of rhombus)
and CD = FC (given)
$\therefore$ BC = FC
$\therefore \hat{B} = \hat{F}_3$ (∠s opp = sides)
$\therefore \hat{F}_3 = 76°$

(3) $\hat{F}_1 = 28°$ (∠s on a str ln)

(4) AB || DC (opp sides of rhombus)
$\therefore \hat{A} = 104°$ (co-int ∠s ; AB || CD)
$\therefore \hat{E}_1 = 104° + 28°$ (ext ∠ of Δ)
$\therefore \hat{E}_1 = 132°$

(d) (1) $\hat{O}_1 = 90°$ (diags of kite)
and $\hat{D}_1 = 45°$ (given)
$\therefore \hat{C}_2 = 45°$ ($\angle$s of Δ)
$\therefore OC = OD$ (sides opp = $\angle$s)

(2) $AC = 17$ cm (given)
but $OD = 12$ cm (diags of kite)
$\therefore AO = 5$ cm
$\hat{O}_2 = 90°$ (diags of kite)
$\therefore AD^2 = AO^2 + OD^2$ (Pythagoras)
$\therefore AD^2 = (5)^2 + (12)^2 = 169$
$\therefore AD = 13$ cm

(e) (1) $\hat{O}_5 = 90°$ (diags of kite)
$KI = KE = 17$ (given)
$\therefore OE^2 = KE^2 - KO^2$ (Pythagoras)
$\therefore OE^2 = (17)^2 - (15)^2 = 64$
$\therefore OE = 8$ m

(2) $\hat{O}_3 + \hat{O}_4 = 90°$ (diags of kite)
$\therefore TE^2 = OT^2 + OE^2$
$\therefore TE^2 = (6)^2 + (8)^2 = 100$
$\therefore TE = 10$ m
but $TP = PE$ (line through midpt $\parallel$ 2nd side)
$\therefore TP = 5$ m

(f) (1) $y = 2x$ ($\angle$s opp main diag of kite)
$\therefore x + 25° + 2x + x + 5° + 2x = 360°$ ($\angle$s of quad)
$\therefore 6x = 330°$
$\therefore x = 55°$
$y = 2x = 2(55°) = 110°$
$z = 180° - 110°$ ($\angle$s on a str ln)
$\therefore z = 70°$

(2) $\hat{O}_1 = 90°$ (diags of kite)
$\therefore x - 8 + x + 90° = 180°$ ($\angle$s of Δ)
$\therefore 2x = 98°$
$\therefore x = 49°$
$y = 25°$ (diags of kite)
$\hat{O}_2 = 90°$ (diags of kite)
$\therefore z = 65°$ ($\angle$s of Δ)

(3) $\hat{P}_1 = \hat{P}_2$ (diags of kite)
$\therefore 2x + 3 = x + 13°$
$\therefore x = 10°$

$\hat{C}_2 = 6x + 1$ ($\angle$s opp = sides)
$\therefore \hat{B}_2 = \hat{C}_2 = 6(10) + 1 = 61°$
$\therefore y = 58°$ ($\angle$s of Δ)

AD $\parallel$ BC (opp sides of rect)
$\therefore \hat{A}_2 = \hat{C}_2 = 61°$ (alt $\angle$s ; AD $\parallel$ BC)
$\therefore z = 61°$

(g) (1) $\text{Area of ABCD} = \frac{1}{2}(\text{AC} \times \text{BD})$
$= \frac{1}{2}(10 \times 6)$
$= 30 \text{ m}^2$

(2) $\text{AO} = 31 - 10 = 21 \text{ cm}$
$\hat{O}_2 = 90°$ (diags of kite)
$\therefore \text{OD}^2 = \text{AD}^2 - \text{AO}^2$ (Pythagoras)
$\therefore \text{OD}^2 = (29)^2 - (21)^2 = 400$
$\therefore \text{OD} = 20 \text{ cm}$
but OD = OB (diags of kite)
$\therefore \text{BD} = 40 \text{ cm}$

$\text{Area of ABCD} = \frac{1}{2}(\text{AC} \times \text{BD}) = \frac{1}{2}(31 \times 40)$
$= 620 \text{ cm}^2$

(h) OS = 4 cm (diags of rhombus)
$\hat{O}_2 = 90°$ (diags of rhombus)
$\therefore \text{PO}^2 = \text{PS}^2 - \text{OS}^2$
$\therefore \text{PO}^2 = (5)^2 - (4)^2 = 9$
$\therefore \text{PO} = 3 \text{ cm}$
$\therefore \text{PR} = 6 \text{ cm}$ (diags of rhombus)
$\text{Area of PQRS} = \frac{1}{2}(\text{QS} \times \text{PR})$
$= \frac{1}{2}(8 \times 6)$
$= 24 \text{ cm}^2$

(i) LN = KM = 3 m (diags of square)
$\text{Area of KLMN} = \frac{1}{2}(\text{LN} \times \text{KM})$
$= \frac{1}{2}(3 \times 3)$
$= 4{,}5 \text{ m}^2$

(j) Area of ABCD $= \frac{1}{2}(BD \times AC)$

$\therefore \frac{1}{2}(24 \times AC) = 612$

$\therefore AC = 51$ cm

$BO = OD = 12$ cm (diag of kite)

$\hat{O}_2 = 90°$ (diag of kite)

$\therefore AO^2 = AD^2 - OD^2$ (Pythagoras)

$\therefore AO^2 = (20)^2 - (12)^2 = 256$

$\therefore AO = 16$ cm

but $AC = 51$ cm

$\therefore OC = 51 - 16 = 35$ cm

$\hat{O}_3 = 90°$ (diag of kite)

$\therefore DC^2 = OD^2 + OC^2$ (Pythagoras)

$\therefore DC^2 = (12)^2 + (35)^2$

$= 1\,369$

$\therefore DC = 37$ cm

EXERCISE 6

(a) $\hat{D} = \hat{B}$ (opp ∠s of parm)

and $\hat{B} = \hat{F}$ (opp ∠s of parm)

$\therefore \hat{D} = \hat{F}$

(b) $PO = OR$ (diags of parm)

and $OR = AT$ (opp sides of parm)

$\therefore PO = AT$

(c) $AE = AC$ (sides opp = ∠s)

but $AC = BD$ (diags of rect)

$\therefore AE = DB$

(d) $\hat{P} = \hat{R}_1$ (∠s opp main diag of kite)

and $\hat{R}_1 = \hat{R}_3$ (vert opp ∠s)

and $\hat{R}_3 = \hat{R}_4$ (diags of rhombus)

$\therefore \hat{P} = \hat{R}_4$

(e) $AO = OD$ (diags of parm)

and $OC = OD$ (sides opp = ∠s)

$\therefore AO = OC$

$\therefore \hat{A}_2 = \hat{C}_2$ (∠s opp = sides)

(f) (1) In ΔABD and ΔCFE:

$AB = FC$ (opp sides of parm)

$\hat{D}_2 = \hat{E}_4$ (alt ∠s ; FG || DB)

$AB \parallel FC$ (opp sides of parm)

$\therefore \hat{A}_1 = \hat{C}_2$ (alt ∠s ; AB || FC)

$\therefore \Delta ABD \equiv \Delta CFE$ (∠∠S)

(2) In ΔAEF and ΔCDB:

$AF = BC$	(opp sides of parm)
$AF \parallel BC$	(opp sides of parm)
$\therefore \hat{A}_2 = \hat{C}_1$	(alt $\angle$s ; $AF \parallel BC$)
$\hat{E}_1 = \hat{E}_3$	(vert opp $\angle$s)
and $\hat{E}_3 = \hat{D}_1$	(corresp $\angle$s ; $FC \parallel DB$)
$\therefore \hat{E}_1 = \hat{D}_1$	
$\therefore \Delta AEF \equiv \Delta CDB$	($\angle\angle$S)

(g) (1) In ΔABG and ΔFDC:

$AG = FC$	(given)
$\hat{A}_1 = \hat{A}_2$	(diag of kite)
and $AE \parallel FD$	(opp sides of parm)
$\therefore \hat{A}_2 = \hat{F}_2$	(corresp $\angle$s ; $AE \parallel FD$)
$\therefore \hat{A}_1 = \hat{F}_2$	
$AB = AE$	(given)
but $AE = FD$	
$\therefore AB = FD$	
$\therefore \Delta ABG \equiv \Delta FDC$	(S$\angle$S)

(2)

$\hat{E}_1 = \hat{B}_1 + \hat{B}_2$	($\angle$s opp main diag of kite)
but $\hat{B}_1 = \hat{D}_1$	($\Delta ABC \equiv \Delta DEF$)
$\therefore \hat{E}_1 = \hat{D}_1 + \hat{B}_2$	

(h) (1)

$AS \parallel QR$	(opp sides of parm)
$PS = SR$	(given)
$\therefore PA = AQ$	(line through midpt $\parallel$ 2nd side)
but $ST = AQ$	(opp sides of parm)
$\therefore PA = ST$	

(2) In ΔPAS and ΔSTR:

$PA = ST$	(proven)
$PS = SR$	(given)
$PQ \parallel ST$	(opp sides of parm)
$\therefore \hat{P} = \hat{S}_1$	(corresp $\angle$s ; $PQ \parallel ST$)
$\therefore \Delta PAS \equiv \Delta STR$	(S$\angle$S)

(i) (1) In ΔOBC and ΔODC:

$EO = OB$	(diags of parm)
and $EO = OD$	(sides opp = $\angle$s)
$\therefore OB = OD$	
$OC = OC$	(common)
$\hat{C}_1 = 90°$	(given)
$\therefore C_2 + C_3 = 90°$	($\angle$s on a str ln)
$\therefore \hat{C}_1 = O\hat{C}D$	
$\therefore \Delta OBC \equiv \Delta ODC$	(RHS)

(2)

$EO = OB$	(diags of parm)
and $BC = DC$	($\Delta OBC \equiv \Delta ODC$)
$\therefore OC \parallel ED$	(midpt thm)

(j) (1) $AC = DB$ (diags of rect)
and $CF = BE$ (given)
$\therefore AC + CF = BD + BE$
$\therefore AF = DE$

(2) In ΔADF and ΔDAE:
$AF = DE$ (proven)
$AD = AD$ (common)
$AO = OD$ (diags of rect)
$\therefore \hat{A}_1 = \hat{D}_1$ ($\angle$s opp = sides)
$\therefore \Delta ADF \equiv \Delta DAE$ (S$\angle$S)

(k) (1) $\hat{A}_1 + \hat{A}_2 = \hat{A}_3 + \hat{A}_4$ (diag of kite)
but $\hat{A}_2 = \hat{A}_3$ (given)
$\therefore \hat{A}_1 = \hat{A}_4$

(2) In ΔABP and ΔADQ:
$\hat{A}_1 = \hat{A}_4$ (proven)
$AB = AD$ (given)
$AP = AQ$ (sides opp = $\angle$s)
$\therefore \Delta ABP \equiv \Delta ADQ$ (S$\angle$S)

(3) $BC = DC$ (given)
but $PB = DQ$ ($\Delta ABP \equiv \Delta ADQ$)
$\therefore BC - BP = DC - DQ$
$\therefore CP = CQ$

(l) In ΔPMN and ΔRAQ:
$PM = AR$ (opp sides of parm)
$PN = QR$ (opp sides of parm)
$MO = QA$ (diags of parm)
and $NO = OQ$ (diags of parm)
$\therefore MO - NO = AO - OQ$
$\therefore MN = AQ$
$\therefore \Delta PMN \equiv \Delta RAQ$ (SSS)

(m) (1) $ED = BF$ (given)
and $DC = CB$ (sides of rhombus)
$\therefore ED + DC = BF + BC$
$\therefore EC = FC$
$\therefore \hat{E} = \hat{F}$ ($\angle$s opp = sides)

(2) In ΔEDQ and ΔFBP:
$ED = BF$ (given)
$\hat{E} = \hat{F}$ (proven)
$\hat{D}_2 = \hat{B}_2$ (opp $\angle$s of rhombus)
but $\hat{D}_1 + \hat{D}_2 = 180°$ ($\angle$s on a str ln)
and $\hat{B}_1 + \hat{B}_2 = 180°$ ($\angle$s on a str ln)
$\therefore \hat{D}_1 = \hat{B}_1$
$\therefore \Delta EDQ \equiv \Delta FBP$ ($\angle\angle$S)

(n) (1) In ΔEOB and ΔFOD:

$EB = DF$	(given)
$OB = OD$	(diags of rhombus)
$\hat{O}_1 = \hat{O}_3 = 90°$	(diags of rhombus)
$\therefore \Delta EOB \equiv \Delta FOD$	(RHS)

(2)

$AO = OC$	(diags of rhombus)
and $EO = OF$	($\Delta EOB \equiv \Delta FOD$)
$\therefore AO - EO = OC - OF$	
$\therefore AE = FC$	

(3) In ΔAEB and ΔCFD:

$AB = DC$	(sides of rhombus)
$AE = FC$	(proven)
$EB = DF$	(given)
$\therefore \Delta AEB \equiv \Delta CFD$	(SSS)

(o) (1) In ΔNQM and ΔLPM:

$NP = QL$	(given)
$\therefore NQ = LP$	
$NM = LM$	(sides of rhombus)
$\hat{N}_1 = \hat{L}_2$	($\angle$s opp = sides)
$\therefore \Delta NQM \equiv \Delta LPM$	(S$\angle$S)

(2)

$\hat{M}_1 + \hat{M}_2 = \hat{M}_2 + \hat{M}_3$	($\Delta NQM \equiv \Delta LPM$)
$\therefore \hat{M}_1 = \hat{M}_3$	

(3)

$\hat{M}_3 = \hat{M}_4$	(diag of kite)
but $\hat{M}_1 = \hat{M}_3$	(proven)
$\therefore \hat{M}_1 = \hat{M}_4$	

(4)

$\hat{T} = \hat{Q}_1$	($\angle$s opp main diag of kite)
but $\hat{Q}_2 = \hat{P}_2$	($\Delta NQM \equiv \Delta LPM$)
but $\hat{P}_1 + \hat{P}_2 = 180°$	($\angle$s on a str ln)
and $\hat{Q}_1 + \hat{Q}_2 = 180°$	($\angle$s on a str ln)
$\therefore \hat{P}_1 = \hat{Q}_1$	
$\therefore \hat{T} = \hat{P}_1$	

EXERCISE 7

(a) (1)

$AD \parallel BC$	(given)
$\hat{D}_1 = 180° - (44° + 26°)$	($\angle$s of Δ)
$\therefore \hat{D}_1 = 110°$	
but $\hat{A} = 110°$	(given)
$\therefore \hat{D}_1 = \hat{A}$	
$\therefore AB \parallel DC$	(corresp $\angle$s =)
$\therefore$ ABCD is a parallelogram	(opp sides of quad $\parallel$)

(2) $AD = DE$ (given)
but $AD = BC$ (opp sides of parm)
$\therefore DE = BC$
and $DE \parallel BC$ (given)
$\therefore$ DBCE is a parallelogram (1 pair of opp sides $\parallel$ and $=$)

(b) (1) $\hat{O}_3 = \hat{R}_2$ (given)
$\hat{T}_4 = 180° - (122° + 16°)$ ($\angle$s of Δ)
$\therefore T_4 = 42°$
and $T_3 = 180° - (40° + 40° + 42°)$ ($\angle$s on a str ln)
$\therefore T_3 = 58°$
but $\hat{Q} = 58°$ (given)
$\therefore T_3 = \hat{Q}$
$\therefore$ OQRT is a parallelogram (opp angles of quad =)

(2) OQ $\parallel$ TR (opp sides of parm)
$\hat{O}_3 = 180° - 58°$ (co-int $\angle$s ; OQ $\parallel$ TR)
$\therefore \hat{O}_3 = 122°$
and $\hat{O}_2 = 360° - (138° + 122°)$ ($\angle$s round a pt)
$\therefore \hat{O}_2 = 100°$
but $\hat{T}_1 + \hat{T}_2 = 80°$ (given)
$\therefore$ PO $\parallel$ UT (co-int $\angle$s suppl)
but PO = UT (given)
$\therefore$ POTU is a parm (1 pair of opp sides $\parallel$ and =)
but $\hat{T}_1 = \hat{T}_2$ (given)
$\therefore$ POTU is a rhombus (parm with diag bisecting $\angle$)

(c) (1) AE = AD (sides opp = $\angle$s)
but AE = BC (given)
$\therefore$ AD = BC
and AB = BD (sides opp = $\angle$s)
but DB = DC (given)
$\therefore$ AB = DC
$\therefore$ ABCD is a parallelogram (opp sides of quad =)

(2) DB $\parallel$ PC (given)
and AP $\parallel$ DC (opp sides of parm)
$\therefore$ BPCD is a parallelogram (opp sides of quad $\parallel$)
but DB = DC (given)
$\therefore$ BPCD is a rhombus (parm with adj sides =)

(d) (1) PA = MR (opp sides of parm)
but PB = CR (given)
$\therefore$ BA = MC
but BA $\parallel$ MC (opp sides of parm)
$\therefore$ BACM is a parallelogram (1 pair of opp sides $\parallel$ and =)
but MB = AB (given)
$\therefore$ BACM is a rhombus (parm with adj sides =)

(2) BA ∥ MC (opp sides of rhombus)
∴ $\hat{A}_1 = \hat{M}_3$ (alt ∠s ; BA ∥ MC)
but $\hat{A}_1 = \hat{C}_1$ (given)
∴ $\hat{M}_3 = \hat{C}_1$
∴ MO ∥ DC (alt ∠s =)
but MD ∥ OC (given)
∴ MOCD is a parallelogram (opp sides of quad ∥)
but $\hat{O}_4 = 90°$ (diags of rhombus)
∴ MOCD is a rectangle (parm with 90°∠)

(e) (1) AD = BC (opp sides of parm)
but AD = AP (given)
∴ AP = BC
but AP ∥ BC (opp sides of parm)
∴ PBCA is a parallelogram (1 pair of opp sides ∥ and =)
AB = DC (opp sides of parm)
and DC = PC (sides opp = ∠s)
∴ PC = AB
∴ PBCA is a rectangle (parm with = diags)

(2) AO = OB (diags of parm)
but AO = CQ (given)
∴ OB = CQ
but OB ∥ CQ (opp sides of parm)
∴ OBQC is a parallelogram (1 pair of opp sides ∥ and =)
but OB = OC (diags of rect)
∴ OBQC is a rhombus (parm with adj sides =)

(f) (1) MO = OA and PO = OR (diags of parm)
but KP = RL (given)
∴ KP + PO = OR + RL
∴ KO = OL
∴ KALM is a parallelogram (diags of quad bisect each other)

(2) In ΔPLM and ΔRKA:
PM = RA (opp sides of parm)
KA = ML (opp sides of parm)
PR + RL = KP + PR (KP = RL given)
∴ KR = PL
∴ ΔPLM ≡ ΔRKA (SSS)

(g) (1) ED = AB (opp sides of parm)
and ED ∥ AB (opp sides of parm)
but ED = DC (given)
∴ AB = DC and AB ∥ DC
∴ ABCD is a parallelogram (1 pair of opp sides ∥ and =)

(2) In ΔADE and ΔADC

$\hat{E} = \hat{C}_1$	(given)
$ED = DC$	(given)
$AE = AC$	(sides opp = $\angle$s)
$\therefore \Delta ADE \equiv \Delta ADC$	(S$\angle$S)

(3)

$\hat{D}_1 + \hat{D}_2 + \hat{D}_3 = 180°$	($\angle$s on a str ln)
but $\hat{D}_1 = \hat{D}_2 + \hat{D}_3$	($\equiv \Delta$s)
$\therefore \hat{D}_2 + \hat{D}_3 = 90°$	
$\therefore$ ABCD is a rectangle	(parm with 90°$\angle$)
$\hat{O}_1 = 90°$	(given)
$\therefore$ ABCD is a rhombus	(parm with $\perp$ diags)
$\therefore$ ABCD is a square	(rhombus and rect)

(h) (1) In ΔAPQ and ΔBRS:

$\hat{Q}_1 = S_1$	(given)
$PQ = SR$	(opp sides of rect)
$\hat{P} = \hat{R} = 90°$	($\angle$ of rect)
$\therefore \Delta APQ \equiv \Delta BRS$	($\angle\angle$S)
$\therefore AQ = SB$	($\equiv \Delta$s)
and $AP = BR$	($\equiv \Delta$s)
but $PS = QR$	(opp sides of rect)
$\therefore AS = QB$	
$\therefore$ AQBS is a parallelogram	(opp sides of quad =)

(2) In ΔPQC and ΔRSD:

$PQ = SR$	(opp sides of rect)
$\hat{Q}_1 = \hat{S}_1$	(given)
$PQ \parallel SR$	(opp sides of rect)
$\therefore \hat{P}_2 = \hat{R}_2$	(alt $\angle$s ; PQ $\parallel$ SR)
$\therefore \Delta PQC \equiv \Delta RSD$	($\angle\angle$S)
$\therefore CQ = SD$	($\equiv \Delta$s)

(i) (1)

$RH = OH$	(sides of rhombus)
but $RB = SO$	(given)
$\therefore RH - RB = OH - SO$	
$\therefore BH = SH$	
In ΔBRM and ΔSOM:	
$RB = OS$	(given)
$MR = MO$	(sides of rhombus)
$\hat{R} = \hat{O}$	(opp $\angle$s of rhombus)
$\therefore \Delta BRM \equiv \Delta SOM$	(S$\angle$S)
$\therefore BM = SM$	($\equiv \Delta$s)
$\therefore$ HSMB is a kite	(adj sides of quad =)

(2)(i) In ΔBHQ and ΔSPQ:

$\hat{Q}_2 = \hat{O}_4 = 90°$	(diags of kite)
$BH = PS$	(given)
$BQ = QS$	(diags of kite)
$\therefore \Delta BHQ \equiv \Delta SPQ$	(RHS)
$\therefore PQ = QH$	($\equiv \Delta$s)
but $BQ = QS$	(diags of kite)
$\therefore$ BHSP is a parallelogram	(diags of quad bisect each other)
but $BH = HS$	(proven)
$\therefore$ BHSP is a rhombus	(parm with adj sides =)

(ii) In ΔBPM and ΔSPM

$BM = SM$	(adj sides of quad =)
$MP = MP$	(common)
$\hat{M}_2 = \hat{M}_3$	(diag of kite)
$\therefore \Delta BPM \equiv \Delta SPM$	(S$\angle$S)

(j) (1)

$RT = EC$	(opp sides of rect)
but $RT = CQ$	(given)
$\therefore EC = CQ$	
$E\hat{C}M = 90°$	($\angle$ of rect)
$\therefore$ TEPQ is a kite	(diag bisects diag at 90°)

(2)

$KL = LQ$	(given)
$MN = NQ$	(given)
$\therefore KM \parallel LN$	(midpt thm)
$\therefore KE \parallel LC$	
$\therefore$ KECL is a trapezium	(1 pair of opp sides of quad $\parallel$)

(3)

$MN = NQ$	(given)
$LC \parallel QP$	(given)
$\therefore MC = CP$	(line through midpt $\parallel$ 2nd side)
but $EC = CQ$	(proven)
$\therefore$ MEPQ is a parallelogram	(diags of quad bisect each other)
but $EP = PQ$	(adj sides of quad =)
$\therefore$ MEPQ is a rhombus	(parm with adj sides =)

CONSOLIDATION AND EXTENSION EXERCISE

(a)

$a = 66°$	($\angle$s opp = sides)
$\therefore b = 48°$	($\angle$s of Δ)
$AP = PB$	(given)
$AO = OC$	(both = $5m$ - given)
$\therefore PO \parallel BC$	(midpt thm)
$\therefore c = 66°$	(corresp $\angle$s ; PO $\parallel$ BC)
$\therefore d = 48°$	($\angle$s of Δ)
$\therefore e = 132°$	($\angle$s on a str ln)
$f = 66°$	(alt $\angle$s ; AP $\parallel$ QO)
$g = 114°$	($\angle$s on a str ln)

(b) PS = SR and PO = OU (given)
$\therefore$ SO $\parallel$ RU (midpt thm)
and $a = 12$ (midpt thm)
PT = TQ (midpt thm)
$\therefore b = 15$ (line through midpt $\parallel$ 2nd side)
$\therefore c = 11$

(midpt thm)

In ΔPQR:
$PQ^2 + PR^2 = RQ^2$
$\therefore 30^2 + d^2 = 34^2$ (Pythagoras)
$\therefore d^2 = 34^2 - 30^2$
$\therefore d^2 = 256$
$\therefore d = 16$

(c) (1) $a = 64°$ (diag of kite)
$b = 2(64) = 128°$ (opp $\angle$s of parm)
AB $\parallel$ DC (opp sides of parm)
$\therefore c = 180° - 128°$ (co-int $\angle$s ; AB $\parallel$ DC)
$\therefore c = 52°$
$d = 52°$
$e = 100° - 64°$ (opp $\angle$s of parm)
$\therefore e = 36°$ (ext $\angle$ of Δ)
$f = 36°$
$g + f + e = 100°$ (diag of kite)
$\therefore g + 36° + 36° = 100°$
$\therefore g = 28°$ (alt $\angle$s ; AB $\parallel$ DC)

(2) $h = 45°$ (diags of square)
EB $\parallel$ DC (opp sides of rhombus)
$i + j = 135°$ (co-int $\angle$s ; EB $\parallel$ DC)
but $i = j$ (diags of rhombus)
$\therefore i = j = 67{,}5°$

(3) AO = OB (diags of rect)
$\therefore l = k$ ($\angle$s opp = sides)
$\therefore l + k + 130° = 180°$ ($\angle$s of Δ)
$\therefore l + k = 50°$
$\therefore l = k = 25°$
$l + m = 90°$ ($\angle$ of rect)
$\therefore m = 90° - 25°$
$\therefore m = 65°$

(d) (1) In ΔAOB and ΔADC:
$\hat{A}_1 = A_2$ (diag of kite)
$A\hat{D}C = 90°$ (given)
and $\hat{O}_2 = 90°$ (diags of kite)
$\therefore \hat{O}_2 = A\hat{D}C$
$\hat{B}_1 = \hat{C}_1$ (3rd $\angle$ of Δ)
$\therefore \Delta AOB \,|||\, \Delta ADC$ ($\angle\angle\angle$)

(2) $\frac{AO}{AD} = \frac{AB}{AC}$ ($\parallel\parallel \Delta$s)

$\therefore \frac{9}{15} = \frac{15}{AC}$ (AD = AB)

$\therefore 9AC = 225$

$\therefore AC = 25$ cm

(3) $OB^2 = AB^2 - AO^2$ (Pythagoras)

$\therefore OB^2 = (15)^2 - (9)^2$

$\therefore OB^2 = 144$

$\therefore OB = 12$ cm

$\therefore DB = 24$cm (diags of kite)

$\text{Area} = \frac{1}{2}(AC \times DB)$

$= \frac{1}{2}(25 \times 24)$

$= 300$ cm^2

(e) (1) $PC^2 = AC^2 - AP^2$ (Pythagoras)

$\therefore PC^2 = (\sqrt{80})^2 - (4)^2$

$\therefore PC^2 = 64$

$\therefore PC = 8$ cm

(2) $AD = DC = x$ (sides of rhombus)

and $PD = 8 - x$

$\therefore AD^2 = AP^2 + PD^2$ (Pythagoras)

$\therefore (x)^2 = (4)^2 + (8 - x)^2$

$\therefore x^2 = 16 + 64 - 16x + x$

$\therefore 16x = 80$

$\therefore x = 5$ m

(3) Area $= b \times \perp h$

$\therefore$ Area $= PC \times AP$

$\therefore$ Area $= 8 \times 4$

$\therefore$ Area $= 32$ m^2

(f) (1)(i) $DO = OB$ (diags of parm)

$OC = \frac{1}{2}(16) = 8$ (diags of parm)

$OQ = OC - QC$

$= 8 - 3$

$= 5$

$\therefore PO = OQ$ (both = 5 units)

$\therefore$ PBQD is a parallelogram. (diags of quad bisect each other)

(ii) $OB = \frac{1}{2}(24) = 12$ (diags of parm)

$OP^2 + OB^2 = 5^2 + 12^2$

$= 169$

$PB^2 = 13^2 = 169$

$\therefore OP^2 + OB^2 = PB^2$

$\therefore \hat{O}_1 = 90°$ (converse Pythagoras)

$\therefore$ PBQD is a rhombus. (parm with $\perp$ diags)

(2) ABCD is a rhombus. (parm with $\perp$ diags)

Shaded area = Area of ABCD − Area of PBQD

$= \frac{1}{2}(AC \times DB) - \frac{1}{2}(PQ \times DB)$

$= \frac{1}{2}(16 \times 24) - \frac{1}{2}(10 \times 24)$

$= 72$ units^2

(g) (1) AB $\parallel$ GC (opp sides of parm)

$\therefore \hat{B}_1 = \hat{G}_2$ (alt $\angle$s ; AB $\parallel$ GC)

GO $\parallel$ FC (opp sides of parm)

$\therefore \hat{G}_2 = \hat{C}_3$ (alt $\angle$s ; GO $\parallel$ FC)

$\therefore \hat{B}_1 = \hat{C}_3$

and $\hat{C}_3 = \hat{D}$ (corresp $\angle$s ; FC $\parallel$ ED)

$\therefore \hat{B}_1 = \hat{D}$

(2) GC = CD and FC $\parallel$ ED (given)

$\therefore$ GF = FE (line through midpt $\parallel$ 2nd side)

and GF = OC (opp sides of parm)

$\therefore$ FE = OC

and OC = AO (diags of parm)

$\therefore$ FE = AO

(3) GC = CD (given)

and GF = FE (proven)

$\therefore$ ED = 2FC (midpt thm)

GB = 2GO (diags of parm)

but GO = FC (opp sides of parm)

$\therefore$ GB = 2FC

$\therefore$ ED = 68 (both = 2FC)

(h) (1) CG = GD and AD $\parallel$ FG (given)

$\therefore$ AF = FC (line through midpt $\parallel$ 2nd side)

$\therefore$ AC = 2FC

but FH = HC (given)

$\therefore$ FC = 2HC

$\therefore$ AC = 2(2HC)

$\therefore$ AC = 4HC

(2) AF = FC (proven)
and EF ∥ BC (given)
∴ AE = EB (line through midpt ∥ 2nd side)
∴ BC = 2EF (midpt thm)
but EF = HG
∴ BC = 2HG

CH = HF and CG = GD (given)
∴ FD = 2HG (midpt thm)

$$ED = EF + FD$$
$$= HG + 2HG$$
$$ED = 3HG$$
$$\therefore HG = \frac{1}{3}ED$$

$$\therefore BC = 2\left(\frac{1}{3}ED\right)$$
$$\therefore BC = \frac{2}{3}ED$$

(i) (1) AB = BC (given)
AP = PG (given)
∴ BP ∥ CG (midpt thm)
BC = CD (given)
and CG ∥ BP (proven)
∴ PO = OD (line through midpt ∥ 2nd side)
OG = OP (sides opp = ∠s)
but OG = 15 cm (given)
∴ OP = 15 cm
∴ DP = 30 cm

(2) CO = x (given)
and OG = 15 cm (given)
∴ CG = $x + 15$
but CG = 2BP (midpt thm)
In ΔBPD:
BC = CD (given)
and PO = OD (proven)
∴ BP = 2CO (midpt thm)
∴ BP = $2x$
∴ $2(2x) = x + 15$
∴ $4x - x = 15$
∴ $x = 5$ cm

(3) In ΔBDR:
BC = CD (given)
and DR ∥ CO (given)
∴ BO = OR (line through midpt ∥ 2nd side)
but PO = OD (proven)
∴ BPRD is a parallelogram (diags of quad bisect each other)

(j) (1) $FE = ED$ (given)
$EB \parallel DC$ (given)
$\therefore FO = OC$ (line through midpt $\parallel$ 2nd side)
$AF = BO$ (opp sides of parm)
and $AF = OE$ (opp sides of parm)
$\therefore BO = OE$
$\therefore$ FBCE is a parallelogram (diags of quad bisect each other)

(2) $EB \parallel DC$ (given)
$FE \parallel BC$ (opp sides of parm)
$\therefore ED \parallel BC$
$\therefore$ BCDE is a parallelogram (opp sides of quad $\parallel$)

(k) (1) $AD = BC$ (given)
$\hat{D}_2 = \hat{B}_2$ (given)
$\therefore AD \parallel BC$ (alt $\angle$s =)
$\therefore$ ABCD is a parallelogram (1 pair of opp sides $\parallel$ and =)

(2) $AB = DC$ (opp sides of parm)
but $PB = DQ$ (given)
$\therefore AP = QC$
but $AP \parallel QC$ (opp sides of parm)
$\therefore$ APCQ is a parallelogram (1 pair of opp sides $\parallel$ and =)

(3) In ΔPBS and ΔQDR:
$PB = DQ$ (given)
$AB \parallel DC$ (opp sides of parm)
$\therefore \hat{B}_1 = \hat{D}_1$ (alt $\angle$s ; AB $\parallel$ DC)
$\hat{P}_2 = \hat{Q}_2$ (opp $\angle$s of parm)
$\therefore \hat{P}_1 = \hat{Q}_1$ ($\angle$s on a str ln)
$\therefore \Delta PBS \equiv \Delta QDR$ ($\angle\angle$S)

(4) $AQ = PC$ (opp sides of parm)
but $PS = RQ$ ($\equiv \Delta$s)
$\therefore AR = SC$
and $AR \parallel SC$ (opp sides of parm)
$\therefore$ ASCR is a parallelogram (1 pair of opp sides $\parallel$ and =)

(l) (1) $AB = OC$ (opp sides of parm)
and $OC = DE$ (opp sides of parm)
$\therefore AB = DE$
but $AB \parallel OC$ (opp sides of parm)
and $OC \parallel DE$ (opp sides of parm)
$\therefore AB \parallel DE$
$\therefore$ AEDB is a parallelogram (1 pair of opp sides $\parallel$ and =)
$AB = AE$ (sides opp = $\angle$s)
$\therefore$ AEDB is a rhombus (parm with adj sides =)

(2) $BC = AO$ (opp sides of parm)
and $AO = OD$ (diags of rhombus)
$\therefore BC = OD$
but $BC \parallel AO$ (opp sides of parm)
$\therefore BC \parallel OD$
$\therefore$ BODC is a parallelogram (1 pair of opp sides $\parallel$ and $=$)
$\hat{O}_5 + \hat{O}_6 = 90°$ (diags of rhombus)
$\therefore$ BODC is a rectangle (parm with $90°\angle$)

(m) (1) In ΔADC and ΔBCD:
$DC = DC$ (common)
$\hat{A}_1 = \hat{B}_1$ (given)
$\hat{C}_3 = \hat{D}_1$ ($\angle$s opp $=$ sides)
$\Delta ADC \equiv \Delta BCD$ ($\angle\angle$S)

(2) $AC = DB$ ($\equiv \Delta$s)
$\therefore$ ABCD is a rectangle (parm with $=$ diags)

(3) $PB \parallel QC$ (opp sides of rect)
$PQ \parallel BC$ (given)
$\therefore$ PBCQ is a parallelogram (opp sides of quad $\parallel$)
$\hat{B}_1 + \hat{B}_2 = 90°$ ($\angle$ of rect)
PBCQ is a rectangle (parm with $90°\angle$)
$\hat{P}_1 + \hat{P}_2 = 90°$ ($\angle$ of rect)
but $\hat{P}_1 = 45°$ (given)
$\therefore \hat{P}_2 = 45°$
$\therefore$ PBCQ is a rhombus (parm with diag bisecting $\angle$)
$\therefore$ PBCQ is a square (rhombus and rect)

(n) (1) In ΔADC:
$AP = PD$ (given)
and $PQ \parallel DC$ (given)
$\therefore AO = OC$ (line through midpt $\parallel$ 2nd side)
In ΔABC:
$AO = OC$ (proven)
and $BQ = QC$ (given)
$\therefore AB \parallel OQ$ (midpt thm)
but $PQ \parallel DC$ (given)
$\therefore AB \parallel DC$
$\therefore$ ABCD is a trapezium (1 pair of opp sides of quad $\parallel$)

(2) $PO = \frac{1}{2}DC$ (midpt thm)
and $OQ = \frac{1}{2}AB$ (midpt thm)
$\therefore PO + OQ = \frac{1}{2}DC + \frac{1}{2}AB$
$\therefore PQ = \frac{1}{2}(AB + DC)$

(3) Area of trapezium $= \frac{1}{2}(AB + DC) \times AD$

$= PQ \times AD$

$= 9 \times 8$

$= 72\ cm^2$